Table 2
Notation.

c	state of charge (SOC)
$\hat{c}$	maximum SOC
D	set of depots; $D = \{1, \dots, d\}$
J	set of trips; $J = \{d+1, \dots, d+n\}$
j	depot, trip or station index
S	set of charging stations; $S = \{d+n+1, \dots, d+n+s\}$
$\tilde{c}_j$	required charge to complete trip j
$\bar{c}_{j,j'}$	required charge for the transit from depot, station, or (the end of) trip j to depot, station, or (the start of) trip j'
e_j	end time of trip j
$\bar{m}_j$	maximum vehicle capacity of depot j
m	number of vehicles
r	constant time required to attach a vehicle to a charger (or to exchange the battery, depending on the context)
s_j	start time of trip j
$t_{j,j'}$	transit time from depot, station, or (the end of) trip j to depot, station, or (the start of) trip j'
$\delta_k(l)$	mapping signifying that vehicle k visits station $\delta_k(l) \in S$ or no station at all ($\delta_k(l) = 0$) after the lth trip of its schedule
$\zeta(j, j', j'', c)$	function giving the maximum SOC at the end of trip (or depot) j' for a vehicle coming from j with SOC c visiting station $j'' \in S$ or no station at all ($j'' = 0$) in-between
$\bar{\zeta}(k, l)$	SOC at the end of the lth trip of vehicle k
$\eta(\tau)$	battery charging function dependent on time τ; $\eta(\tau)$ is the SOC if an empty battery is charged for time τ
τ	amount of time
$\bar{\tau}(j, j', j'', c)$	time a vehicles spends charging at station j'' while transiting from j with SOC c to j'
ω_k	ordered set denoting which depots are to be visited and trips are to be performed by vehicle k in what order

cle and operator costs tend to dominate the detour costs due to the relatively short driving distances (Emde et al., 2018; Golz et al., 2012).

3.1. Formal problem description

We base the definition of the EVSP-MD-FS on the notation summarized in Table 2. Let $D = \{1, \dots, d\}$ be the set of depots, let $J = \{d+1, \dots, d+n\}$ be the set of trips to be processed during the planning horizon and let $S = \{d+n+1, \dots, d+n+s\}$ be the set of charging stations (called "stations" in the following). Note that we introduce offset indices to ease notation in the following definitions and models. Generally speaking, the locations of stations do not have to coincide with the depots or the start or end points of the trips, but can of course correspond to these locations. In fact, the former is the case for most of the in-plant milk-run logistic systems we observed. Each trip consists of a vehicle visiting multiple locations (i.e., assembly cells) in a given path. The exact path the vehicle takes is immaterial for the EVSP-MD-FS; we assume that this issue has been decided in a previous step. Let s_j be the start time of trip j, i.e., the time when the vehicle arrives at the trip's first location, and let e_j be the end time of trip j, i.e., the time when the vehicle departs from the trip's last location. For notational convenience, we also define $s_j = \infty$ and $e_j = 0$, $\forall j \in D \cup S$, which implies that depots and stations are always available. Without loss of generality, we assume that $s_j \le s_{j'}$ holds, $\forall j, j' \in J : j < j'$, i.e., the trips are indexed according to their starting time in ascending order. Moreover, let $t_{j,j'}$, $\forall j, j' \in D \cup S \cup J : j \neq j'$, be the transit time of a vehicle from depot, station, or the end of trip j to depot, station, or the starting point of trip j'. For transit times, we assume that the triangle inequality holds in the sense that $t_{j,j'} \le t_{j,j''} + \max\{e_{j''} - s_{j''}, 0\} + t_{j'',j'}$, $\forall j, j', j'' \in J \cup D \cup S$, where $\max\{e_{j''} - s_{j''}, 0\}$ is the trip's execution time if $j'' \in J$, and zero if $j'' \in D \cup S$. Let the battery capacity, i.e., maximum SOC, of the vehicle be $\hat{c}$, and the charge required to complete trip $j \in J$ be $\tilde{c}_j$, where we assume $\tilde{c}_j \le \hat{c}$ applies, $\forall j \in J$. Note that the required charge $\tilde{c}_j$ may depend on the total distance covered, the number of stopovers, the carried load, the terrain (e.g., steep slopes), and other factors. We assume that these values have been determined beforehand and are given. To ease notation, we further define $\tilde{c}_j = 0$ and $\bar{c}_{j,j} = 0$, $\forall j \in D \cup S$. Let $\bar{c}_{j,j'}$ be the charge required to drive from depot, station or the end of trip j to depot, station, or the starting point of trip j'. As for travel times, we assume that the triangle inequality also holds for battery requirements in the sense that $\bar{c}_{j,j'} \le \bar{c}_{j,j''} + \tilde{c}_{j''} + \bar{c}_{j'',j'}$, $\forall j, j', j'' \in J \cup D \cup S$. We assume that charging is only possible at stations. If vehicles should be able to charge at depots or at the start or end points of trips, we can add stations that correspond to the respective locations.

Let $\eta(\tau)$ denote the charging function over time τ, i.e, the SOC an empty battery would have after being charged for a time of τ. We assume that $\eta(\tau)$ is either a constant function, a linear function or a non-linear function, which, in the latter two cases, is capped at $\hat{c}$, i.e., when the battery is full. Moreover, we assume that a vehicle arriving at a station requires a time of r before charging can start, where, depending on the assumed charging technology, r is the time required to swap the battery or to connect the battery to the charger. This enables us to account for the two most commonly encountered charging technologies in practice, namely battery swapping and charging via power cable with a realistic non-linear increase of the SOC. Furthermore, we assume the charging function is equal for all vehicles and stations and that vehicles are always charged whenever they spend sufficient time at a charging station. For a vehicle charging for a time of τ beginning from a SOC $c \ge 0$, the resulting SOC is given as $\eta\left(\eta^{-1}(c) + \tau\right)$, where $\eta^{-1}(c)$ is the inverse function of $\eta(\tau)$ and states the time τ required to charge an empty battery to SOC c. For the case of $\eta(\tau)$ being a constant function, we define $\eta^{-1}(c) = 0$. To further ease notation, we define $\eta^{-1}(c) = 0$, for $c < 0$, and $\eta^{-1}(c) = \infty$, for $c > \hat{c}$, as well as $\eta(\tau) = 0$, for $\tau < 0$. This makes it easy to accommodate non-linear charging.

A solution for EVSP-MD-FS then consists of

- a collection $\{\omega_1, \dots, \omega_m\}$ of ordered sets ω_k, denoting which depots are to be visited and which trips are to be performed by each vehicle $k = 1, \dots, m$ in what order,
- and m mappings $\delta_k : \{1, \dots, |\omega_k| - 1\} \to j$, for $k = 1, \dots, m$, where $\delta_k(l) = j$ specifies the deadheading route of vehicle k between the lth trip (or depot for $l = 1$) and the $(l+1)$th trip (or depot for $l = |\omega_k| - 1$). Two general types of deadheading routes are possible: If $j = 0$, the vehicle drives directly form trip l to $l+1$ without any charging break. Otherwise, if $j \in S$, the vehicle visits station j after the lth trip (or depot) to recharge its battery before heading for the $(l+1)$th trip (or depot).

Let $\omega_k(l)$ refer to the lth element in the ordered set. For all vehicles $k = 1, \ldots, m$, it must hold that $\omega_k(1) \in D$ and $\omega_k(|\omega_k|) \in D$, since each vehicle departs from a depot initially and returns to one at the end. Furthermore, all trips need to be processed exactly once, i.e., $\bigcup_{k=1}^{m} \bigcup_{l=2}^{|\omega_k|-1} \{\omega_k(l)\} = J$ and $\omega_k(l) \neq \omega_{k'}(l')$, $\forall k, k' = 1, \ldots, m; l, l' = 2, \ldots, |\omega_k| - 1 : k \neq k' \vee l \neq l'$ must hold.

Note that by these definitions, vehicles are also allowed (but not required) to visit a charging station right after leaving and before returning to a depot. Moreover, our definition of δ_k imposes that vehicles visit at most one station between consecutively executed service trips. This forecloses the possibility of chains of charging, which would be especially useful if distances between service trips are large compared to the vehicles' battery reach, such that multiple charges are required to travel between consecutive service trips. Since distances are comparatively small in intralogistic transportation systems, visiting multiple charging stations in direct succession is all but unheard of, however.

We say a solution to an instance of EVSP-MD-FS is feasible if it is *feasibly executable*. I.e., for every vehicle, the battery's SOC must never be negative and the executed route must not violate the time constraints. For a formal definition, see Section 3.2.

Moreover, while somewhat uncommon in the in-plant transportation systems that motivated this paper, there are additional constraints that planners may reasonably want to consider. First, due to space constraints, each depot $j \in D$ could be required to host no more than $\bar{m}_j$ vehicles, where $\bar{m}_j = n$ if no limit exists. Second, vehicles could be required to return to their depot of initial departure at the end of their schedule. In the following, we discuss them as optional extensions of the basic EVSP-MD-FS problem. Specifically, $|\{k \in \{1, \ldots, m\} : \omega_k(1) = j\}| \leq \bar{m}_j \wedge |\{k \in \{1, \ldots, m\} : \omega_k(|\omega_k|) = j\}| \leq \bar{m}_j$, $\forall j \in D$, must hold to account for the limited capacity of the depots. Finally, if vehicles are required to return to their depot of initial departure, it must hold that $\omega_k(1) = \omega_k(|\omega_k|)$, $\forall k \in \{1, \ldots, m\}$.

The objective of EVSP-MD-FS is to find a feasible schedule that minimizes the number of vehicles m. Hence, we minimize the number of vehicles m to which trips are assigned.

3.2. Feasibly executable *solutions*

In order to determine whether a solution is *feasibly executable*, we must determine the recharged battery during charging breaks and, hence, their duration. Each vehicle can charge its battery whenever it visits a station, i.e., whenever $\delta_k(l) \in S$. We can preempt the decision on the charging breaks' duration by taking the following proposition into account.

Proposition 3.1. *Given the trips (and depots) ω_k to be executed (or visited) by vehicle k, the policy of charging the vehicle at every charging station it visits for as long as the trip sequence ω_k permits creates a feasible solution, if any exists. I.e., using this charging policy, the vehicle reaches each trip (and depot) $j \in \omega_k$ with the maximum SOC possible given ω_k.*

Proof. The proof is by an interchange argument. Assume that in some feasible solution, vehicle k reaches trip $j \in \omega_k : j \neq \omega_k(|\omega_k|)$ with SOC c' and the consecutive trip (or depot) j' with c'', where it executes the deadheading route j'' in-between. We refer to this as case one. Alternatively, assume a second case, where j is reached with SOC $c''' = c' + \Delta c$ and j' with SOC c'''', where Δc is a positive amount of charge. In the following, we prove that $c'''' \geq c''$ is guaranteed to hold true.

In the second case, the vehicle can execute the exact same deadheading route as in the first case. If $j'' = 0$, it follows that $c'' = c' - \bar{c}_{j,j'}$ and $c'''' = c''' - \bar{c}_{j,j'} = c' + \Delta c - \bar{c}_{j,j'}$, hence $c'''' > c''$ applies. Else, if $j'' \in S$, in case one, the vehicle reaches j'' with SOC $c' - \bar{c}_{j,j''}$. Let τ' denote the time the vehicle charges at station j'', such that it has SOC $\eta\left(\eta^{-1}\left(c' - \bar{c}_{j,j''}\right) + \tau'\right)$ when leaving j''. Consequently, $c'' = \eta\left(\eta^{-1}\left(c' - \bar{c}_{j,j''}\right) + \tau'\right) - \bar{c}_{j'',j'}$ follows. In the second case, the vehicle reaches station j'' with $c' + \Delta c - \bar{c}_{j,j''}$ and can charge for the exact same duration τ' as in the first case. Consequently, it follows that $c'''' = \eta\left(\eta^{-1}\left(c' + \Delta c - \bar{c}_{j,j''}\right) + \tau'\right) - \bar{c}_{j'',j'}$. Since $\eta^{-1}(c)$ is an increasing function of c and $\eta(\tau)$ is an increasing function of τ, $c'''' \geq c''$ is guaranteed to hold true.

The same reasoning applies for all consecutive trips in ω_k until the final depot $\omega_k(|\omega_k|)$ is reached. Hence, if there exists a feasible solution, there also exists a feasible solution where the remaining charge when reaching each $j \in \omega_k$ is maximized, that is, where charging breaks are never cut short before the vehicle must depart for its next trip. □

Therefore, at each visited station, the vehicle should charge for the maximum possible duration that still allows to timely reach the consecutive trip or depot. Given $\delta_k(l) = j''$ with $j'' \in S$, let $\omega_k(l) = j$ and $\omega_k(l+1) = j'$ be two consecutively executed trips or one of them be a depot. Furthermore, let c be the SOC when the vehicle leaves from trip (or depot) j. The optimal time $\bar{\tau}(j, j', j'', c)$ the vehicle spends charging at station j'' then follows as

$$\bar{\tau}\left(j, j', j'', c\right) = s_{j'} - e_j - t_{j,j''} - t_{j'',j'} - r, \tag{1}$$

which is comprised of the available time between the start of trip j' and the end of trip j minus the travel time to and from station j'' minus the plug-in/ battery swap time r.

Given two consecutively executed trips $\omega_k(l) = j$ and $\omega_k(l+1) = j'$ (where either might be a depot instead) and $\delta_k(l) = j''$ (with $j'' \in S \cup \{0\}$) indicating the deadheading route in-between, the maximal SOC at the end of trip (or depot) j' can be calculated as

$$\zeta\left(j, j', j'', c\right) = \begin{cases} c - \bar{c}_{j,j'} - \tilde{c}_{j'} & \text{if } j'' = 0 \wedge e_j + t_{j,j'} \leq s_{j'} \\ \eta\left(\eta^{-1}\left(c - \bar{c}_{j,j''}\right) + \bar{\tau}\left(j, j', j'', c\right)\right) & \text{if } j'' \in S \wedge \bar{c}_{j,j''} \leq c \\ \quad -\bar{c}_{j'',j'} - \tilde{c}_{j'} & \quad \wedge \bar{\tau}\left(j, j', j'', c\right) \geq 0 \\ -1 & \text{otherwise} \end{cases}. \tag{2}$$

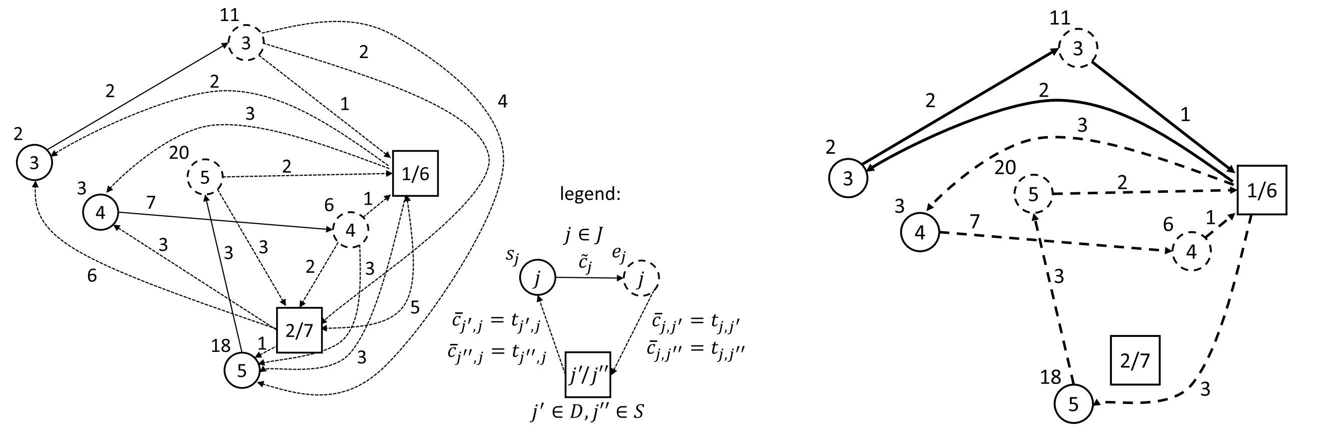
If no station is visited at the deadheading route, the first case applies. Here, the SOC at the end of trip j' is the SOC at the end of trip j minus the battery charge consumed for the transit from j to j' and for the execution of j'. If a station is visited in-between j and j', the second case applies. Here, the vehicle reaches station j'' with SOC $c - \bar{c}_{j,j''}$ and charges for a duration of $\bar{\tau}\left(j, j', j'', c\right)$, which results in a SOC of $\eta\left(\eta^{-1}\left(c - \bar{c}_{j,j''}\right) + \bar{\tau}\left(j, j', j'', c\right)\right)$ at the end of the charging procedure. Afterwards, trip j' must be reached and executed, which reduces the SOC at the end of trip j' by $\bar{c}_{j'',j'} + \tilde{c}_{j'}$. If either the charge c is not sufficient to reach the station j'' or there is not enough time, $\zeta\left(j, j', j'', c\right)$ assumes a value < 0 to indicate infeasibility.

We define $\bar{\zeta}(k, l)$ as the SOC of vehicle k at the end of the lth trip or depot, where $\bar{\zeta}(k, |\omega_k|)$ denotes the SOC when vehicle k reaches the final depot. For $l \geq 2$, it can be recursively calculated as

$$\bar{\zeta}(k, l) = \zeta\left(\omega_k(l-1), \omega_k(l), \delta_k(l-1), \bar{\zeta}(k, l-1)\right),$$

with $\bar{\zeta}(k, 1) = \hat{c}$ for $l = 1$.

Based on these definitions, a solution to EVSP-MD-FS is *feasibly executable* if and only if $\zeta(k, l) \geq 0$, $\forall k \in \{1, \ldots, m\}, l \in \{1, \ldots, |\omega_k|\}$. I.e., for every vehicle, the battery's SOC must never be negative and the executed route must not violate the time constraints (since, otherwise, $\zeta\left(j, j', j'', c\right)$ and, hence, $\bar{\zeta}(k, l)$, would assume the value -1).



(a) Example problem data. (b) Optimal solution in the example.

Fig. 2. Example data and solution.

Example. Consider the example problem depicted in Fig. 2a, consisting of $n = 3$ trips and $d = s = 2$ depots that are also charging stations. Let the battery capacity of the vehicles be $\hat{c} = 11$, the charge function be linear $\eta(\tau) = \tau$ (with the cap $\eta(\tau) = 11$, for $\tau \geq 11$) and the constant charge time be $r = 0$. Furthermore, let $\bar{m}_j = n$, $\forall j \in D$, and let the vehicles be not required to return to their initial depot. Note that in this example, we assume that all driving times ($t_{j,j'}$) are equal to the energy expenditures ($\bar{c}_{j,j'}$), which generally does not need to be the case. The optimal solution is depicted in Fig. 2b, corresponding to $m = 2$ vehicles being used. Solid arrows denote the route of vehicle 1, dotted arrows the route of vehicle 2. Vehicle 1 processes only trip 3 (i.e., $\omega_1 = \{1, 3, 1\}$, $\delta_1(1) = 0$, $\delta_1(2) = 0$, $\bar{\zeta}(1, 1) = 11$, $\bar{\zeta}(1, 2) = 7$, $\bar{\zeta}(1, 3) = 6$). Vehicle 2 processes trips 4 and 5 (in that order), returning to, first, station 6 – where it charges its battery for a time of $\bar{\tau}(4, 5, 6, 1) = 18 - 6 - 1 - 3 - 0 = 8$ – and, second, depot 1 after the trips (i.e., $\omega_2 = \{1, 4, 5, 1\}$, $\delta_2(1) = 0$, $\delta_2(2) = 6$, $\delta_2(3) = 0$, $\bar{\zeta}(2, 1) = 11$, $\bar{\zeta}(2, 2) = 1$, $\bar{\zeta}(2, 3) = 2$, $\bar{\zeta}(2, 4) = 0$).

3.3. Complexity

Concerning the time complexity of EVSP-MD-FS, classic (single-depot) non-electric vehicle scheduling is well-known to be tractable (Saha, 1970), as it can be reduced to finding a minimum cost perfect matching in a bipartite graph (Bertossi, Carraresi, & Gallo, 1987). This result also holds for the special single-depot case of EVSP-MD-FS where the battery capacity is not a limiting factor ($\hat{c} = \infty$). In this case, charging breaks need not be considered and the driving time t to/from the depot is immaterial. The problem then is to assign the given trips to as few vehicles as possible such that no trips overlap, which is equivalent to classic vehicle scheduling with the objective of minimizing the fleet size.

However, as soon as the battery capacity becomes a limiting factor, single-depot electric vehicle scheduling with finite battery capacity and the objective of minimizing the fleet size becomes intractable, as is proven by Emde et al. (2018, Proposition 3) by a reduction from bin packing. Obviously, this result extends to the multi-depot case as considered in EVSP-MD-FS. Finally, the result also holds if additional constraints are considered, since they merely generalize the basic EVSP-MD-FS problem.

3.4. Feasible deadheading routes

By the definitions in Section 3.1, a solution to EVSP-MD-FS consists of two distinctive parts. The ordered subsets $\omega_1, \dots, \omega_m$ denote which depots are to be visited and which trips to be executed by which vehicle in which sequence. The mappings $\delta_1, \dots, \delta_m$ indicate the deadheading route in-between the trips and depots for all vehicles $1, \dots, m$. Given only ω_k, the respective feasible deadheading routes δ_k can be determined efficiently as follows.

Let $\omega_k(l) = j$ and $\omega_k(l+1) = j'$ be two consecutively executed trips or either be a depot. Furthermore, let c be the SOC when the vehicle leaves from trip (or depot) j. By Proposition 3.1, provided that any feasible solution exists at all for a given ω_k, there is always a feasible solution that reaches j' with the highest possible SOC and, consequently, completes j' with the maximum possible SOC (since $\tilde{c}_{j'}$ is constant). The latter SOC is given by $\zeta(j, j', j'', c)$ (cf., Eq. (2)), which beyond j, j' and c is solely dependent on $\delta_k(l) = j''$, the deadheading route between j and j'. Hence, it follows that

$$\delta^*_k(l) = \underset{j'' \in S \cup \{0\}}{\arg\max} \left\{ \zeta\left(j, j', j'', c\right) \right\}$$

is guaranteed to be feasible (if feasibility is attainable at all). Consequently, the maximum and therefore feasible charge at the end of trip j' is given as

$$\zeta^*\left(j, j', c\right) = \max_{j'' \in S \cup \{0\}} \left\{ \zeta\left(j, j', j'', c\right) \right\}.$$

4. IP model

To enable the use of default solvers, we propose an IP model based on the notation given in Table 3. We incorporate non-linear battery charging by discretizing the SOC. I.e., the SOC can only assume discrete values $c \in C$. We note that the discretization is only required for the IP model; the solution procedure presented in Section 5 does not rely on discretization. For linear and constant-time charging, the discrete SOC is exact, as long as we assume that all problem parameters, i.e., travel times and required battery charges, are integer. For non-linear charging, using discrete SOC is an approximation. By scaling the problem parameters accordingly, we can, however, approximate the actual SOC to any desired precision.

We introduce binary variables $\beta_{j,c}$ to track the discrete SOC $c \in C$ at the end e_j of each trip $j \in J$ and when a vehicle reaches the final depot $j \in D$. Note that the number of variables $\beta_{j,c}$ grows pseudo-polynomially with $O(\hat{c} \cdot n)$. Moreover, we define set $N = \{(j, c) \in (J \cup D) \times C \mid c \neq \lfloor \zeta^*\left(j', j, c'\right) \rfloor, \forall j' \in J \cup D, c' \in C\}$ as the set of pairs (j, c) indicating that trip j is never completed with SOC c in the optimal solution. Note that set N can be determined in $O((n+d)^2 \cdot s \cdot \hat{c})$, by simply finding all combinations $j, j' \in J \cup D$: $j < j'$, $c \in C$, where $\max_{j'' \in S \cup \{0\}} \left\{ \zeta\left(j, j', j'', c\right) \right\} = \zeta^*\left(j, j', c\right) < 0$.

Table 3
Notation for the IP model.

C	set of discrete SOC values; index c; $C=\{0,\dots,\hat{c}\}$; a value of c' refers to $\frac{c'}{\hat{c}}$ of the maximum SOC
$\upsilon_{j,j'}$	binary variable: 1, if trip j is associated with depot j', else 0
$x_{j,j'}$	binary variable: 1, if a vehicle transits from trip or depot j to trip or depot j' (with possible detours for charging in-between), else 0
$\beta_{j,c}$	binary variable: 1, if trip j is completed (or depot j is reached) with SOC c

Using these definitions, Eqs. (3) to (13) define an IP model for the case that vehicles are not required to return to their initial depot at the end of their schedules.

$$\text{[EVSP-MD-FS] Minimize } F(\boldsymbol{x},\boldsymbol{\beta})=\sum_{j\in D}\sum_{j'\in J}x_{j,j'} \tag{3}$$

subject to

$$\sum_{\substack{j'\in J\cup D:\\ j'>j\vee j'\in D}} x_{j,j'}=1 \qquad \forall j\in J \tag{4}$$

$$\sum_{\substack{j\in J\cup D:\\ j<j'}} x_{j,j'}=1 \qquad \forall j'\in J \tag{5}$$

$$\sum_{\substack{c\in C:\\ (j,c)\notin N}} \beta_{j,c}=1 \qquad \forall j\in J \tag{6}$$

$$0\geq x_{j,j'}+\beta_{j,c}-1 \qquad \forall j\in J, j'\in J\cup D: j<j'\vee j'\in D,\; c\in C:(j,c)\notin N\wedge\lfloor\zeta^*(j,j',c)\rfloor<0 \tag{7}$$

$$\beta_{j',\lfloor\zeta^*(j,j',c)\rfloor}\geq x_{j,j'}+\beta_{j,c}-1 \qquad \forall j\in J, j'\in J\cup D: j<j'\vee j'\in D,\; c\in C:(j,c)\notin N\wedge\lfloor\zeta^*(j,j',c)\rfloor\geq 0 \tag{8}$$

$$\beta_{j',\lfloor\zeta^*(j,j',\hat{c})\rfloor}\geq x_{j,j'} \qquad \forall j\in D, j'\in J:\lfloor\zeta^*(j,j',\hat{c})\rfloor\geq 0 \tag{9}$$

$$\sum_{j\in J}x_{j',j}\leq\bar{m}_{j'} \qquad \forall j'\in D \tag{10}$$

$$\sum_{j\in J'}x_{j,j'}\leq\bar{m}_{j'} \qquad \forall j'\in D \tag{11}$$

$$x_{j,j'}\in\{0,1\} \qquad \forall j,j'\in J\cup D:(j,j'\in J\wedge j<j')\vee(j\in J\wedge j'\in D)\vee(j\in D\wedge j'\in J\wedge\lfloor\zeta^*(j,j',\hat{c})\rfloor\geq 0) \tag{12}$$

$$\beta_{j,c}\in\{0,1\} \qquad \forall j\in J\cup D, c\in C:(j,c)\notin N \tag{13}$$

Objective function (3) minimizes the number of required vehicles. Constraints (4) and (5) enforce that every trip has exactly one preceding and one succeeding trip or depot. Constraints (6) make sure that each trip is finished with exactly one non-negative SOC.

Constraints (7) to (9) ensure consistent SOC values. Inequalities (7) prohibit the execution of trip j' after trip j has been completed with SOC c if the SOC would be insufficient for completing the trip or time constraints would be violated (i.e., $\lfloor\zeta^*(j,j',c)\rfloor<0$). Likewise, if the SOC is sufficient and no time constraints are violated (i.e., $\lfloor\zeta^*(j,j',c)\rfloor\geq 0$), Inequalities (8) set the SOC at the end of trip j' to the appropriate value. Constraints (9) determine the SOC of the trips that are initially executed after the vehicles leave the depots.

Constraints (10) and (11) enforce limits on the number of vehicles per depot. Finally, Constraints (12) and (13) define the domains of the decision variables. Note that – in order to reduce the search space – we omit $\beta_{j,c}$ for $(j,c)\in N$, since the states can never be feasibly reached.

Moreover, to formulate the IP model, we note that determining $\zeta^*(j,j',c)$ requires a pseudo-polynomial amount $O((n+d)^2\cdot s\cdot\hat{c})$ of pre-calculations. Furthermore, the optimal deadheading routes need to be determined from the IP's solution in a $O((n+d)\cdot s)$ post-processing procedure (cf., Section 3.4).

To enforce the optional constraint that each vehicle return to the depot of its initial departure, we introduce the binary variables $\upsilon_{j,j'}$ that are 1 if trip j is associated with depot j', and 0 otherwise. Using these variables, we extend the IP with the following equations:

$$\sum_{j'\in D}\upsilon_{j,j'}=1 \qquad \forall j\in J \tag{14}$$

$$\upsilon_{j',j''}\geq x_{j,j'}+\upsilon_{j,j''}-1 \qquad \forall j,j'\in J:j<j', j''\in D \tag{15}$$

$$\upsilon_{j,j'}\geq x_{j',j} \qquad \forall j\in J, j'\in D \tag{16}$$

$$\upsilon_{j,j'}\geq x_{j,j'} \qquad \forall j\in J, j'\in D \tag{17}$$

$$\upsilon_{j,j'}\in\{0,1\} \qquad \forall j\in J, j'\in D \tag{18}$$

Constraints (14) make sure that every trip $j\in J$ is associated with exactly one depot. Inequalities (15) force each pair of consecutively executed trips, j and j', to be associated with the same depot. Constraints (16) force the trip executed directly after leaving a depot to be associated with the respective depot. Likewise, Constraints (17) associate each final trip before returning to a depot with the respective depot. In conjunction, these constraints enforce each tour to start from and end at the same depot.

5. Branch-and-check

Our computational tests reveal that a default solver using the IP model from Section 4 performs subpar for instances with $n\geq 100$ (see Section 6.2). We therefore propose a decomposition scheme based on branch-and-check (BCH, Beck, 2010; Thorsteinsson, 2001).

The general idea of BCH is to split a complicated problem into two parts, which are individually easier to solve: a mixed-integer programming master model, which is a relaxation of the original problem, and a subproblem, which encodes the remaining variables and/or constraints, albeit not necessarily in the form of a MIP. The master model is solved using classic branch-and-bound methods. Periodically, when it is advantageous, the subproblem is solved at a node of the branch-and-bound tree to check the feasibility of the current master solution and/ or determine its exact objective value. This information is then communicated to the overarching branch-and-bound process, usually via cuts. The search terminates when there are no more unfathomed nodes in the enumeration tree left to explore. The best found solution is optimal.